

Paul V. Ingram III

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EDUCATION

- M.S. in Computational Data Science** - *University of California, Riverside* Sept. 2026
- B.S. in Computer Science** - *University of California, Riverside* Sept. 2024 - Jun. 2026
- **Overall GPA:** 3.81 / 4.0

TECHNICAL SKILLS

Programming Languages: C/C++, CUDA, Java, Python, Lisp (Scheme), Assembly, Rust
Web Development: JavaScript, HTML, CSS, Next.js, TailwindCSS
Python Libraries: pandas, seaborn, NumPy, Matplotlib, SciPy, ROS, LangChain, LangGraph, discord.py, aiosqlite
AI & APIs: Anthropic Claude, LLM prompt engineering, web search tool use, structured output
DevOps & Tools: Git, GitHub Actions, Polygon for Codeforces, Visual Studio Code, SQLite

PROJECTS

- Senior Design: OS Kernel Exploitation** | *C, Linux, Android*
- Ported an Android kernel **privilege escalation** exploit (**CVE-2020-0041**) to a custom **QEMU** environment
 - Built a **4-stage exploit chain** bypassing **ASLR** to achieve **arbitrary kernel read/write** and escalate to **root**
 - Wrote a **100-line C binder wire protocol** to replace Android IPC, enabling the exploit on standard Linux
- FlashAttention CUDA Kernel** | *CUDA, C++, PyTorch, libtorch* [GitHub](#)
- Implemented **FlashAttention** forward pass as a **CUDA kernel** reducing space complexity from $O(N^2)$ to $O(Nd)$
 - Used **171x less memory** than naive PyTorch
 - Adopted **FlashAttention-2** loop ordering to parallelize over query tiles with no cross-block synchronization
- AI for Robotics Final Project** | *Python, ROS, RViz, Matplotlib* [GitHub](#)
- Simulated **autonomous path planning** and **SLAM** mapping in **ROS** using **Python**
 - **Dijkstra-based** planner over occupancy grids with **RViz** waypoint selection for interactive route control
 - Benchmarked path planning algorithms with **Matplotlib** to compare **runtime and space complexity**
- Data Science Mini Project** | *Python, pandas, Matplotlib* [GitHub](#)
- Cleaned and aggregated public survey data with **Python** and **pandas** for **exploratory data analysis**
 - Conducted **chi-square** and **Pearson correlation** analysis to quantify relationships between variables
 - Identified time spent with family as the strongest predictor of student stress levels ($p < 0.05$)

EXPERIENCE

- Undergraduate Research Assistant** - *University of California, Riverside* Jan. 2026 - Present
- Reproduced Android application **CVEs** in user space by analyzing **vulnerability reports** and executing publicly available **proof-of-concept (PoC) exploits**
 - Designed and implemented a **data collection workflow** to scrape and curate **33 Android CVEs** from public **vulnerability databases**
 - Assisted in evaluating **automated vulnerability analysis** tooling by importing target **APKs** and curated **CVE datasets**

LEADERSHIP

- Grader** - *University of California, Riverside* Jan. 2026 - Present
- CS 105 (Data Analysis Methods): Helping with grading labs and quizzes for a class of ~100 students
 - CS 111 (Discrete Structures): Helping with grading assignments and quizzes for a class of over 150 students
- President – ACM Student Chapter** - *Riverside Community College* Aug. 2022 - Jun. 2024
- Led competitive programming workshops and coordinated technical programming events
 - Conducted outreach by developing introductory competitive programming activities and delivering guest lectures to high school students